STATISTICS

Statistics Cheat Sheet

What interviewers ask, one line each.

@thedataguy16

Swipe →

The basics

- **Median over mean**
  When data is skewed
- **68 · 95 · 99.7**
  Within 1, 2, 3 standard deviations
- **Central Limit Theorem**
  Sample means look normal

Hypothesis testing

- **p-value**
  Chance of this result if nothing changed
- **Below 0.05**
  Usually significant
- **Type I**
  False alarm
- **Type II**
  Missed a real effect

Which test?

- **2 group means**
  `t-test`
- **3+ group means**
  `ANOVA`
- **2 categories**
  `Chi-square`
- **2 conversion rates**
  `z-test`

A/B test in 5 steps

- Pick one metric
- Fix sample size first
- Split users randomly
- Run full weeks, no peeking
- Check significance, then decide